

Worked example — not a real business

The company in this report is fictional and every input is illustrative. The report shows what the platform produces; it is not advice or a recommendation.

① Decision report

Sample report — a fictional company, illustrative figures

Indicative — 7 business inputs: 6 supplied or fitted, 1 estimated; plus 2 fixed template defaults

Decision

Raise prices 8% vs hold — Kestrel Software

Decision period 24 months

Model result

Raise prices 8%

Better cumulative net cash flow over 24 months than **Hold prices**, with both options tested against the same simulated conditions: a like-for-like comparison rather than two separate forecasts.

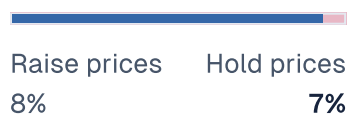
Unsettled Volume loss could reverse this result without leaving the range entered for it, and that range is an estimate with no supporting evidence.

Probability

93%

of valid simulated scenarios in which Raise prices 8% came out ahead

Scenario comparison



Mean advantage

\$231,759

Mean cumulative net cash flow over 24 months

Raise prices 8% **\$1,101,972**

Hold prices **\$870,213**

against Hold prices · computed before rounding, so it can sit \$1 from the difference of the two means above

Downside

\$586,973

P10 for Raise prices 8%: nine in ten simulated scenarios land better than this.

Hold prices **\$395,745**

Downside comparison

At the downside end (P10), Raise prices 8% returns **\$586,973** and Hold prices returns **\$395,745**. Nine in ten simulated scenarios land better than this for each option. Compared scenario by scenario, Raise prices 8% is at least \$31,901 ahead in nine scenarios out of ten.

Cash floor

Neither option runs out of cash in any simulated scenario. Cash is not the deciding factor here.

Before you read the result

- Model constraints: 4 constraint(s) required for “raise prices, or hold” decisions were applied automatically. A constraint you declare on the same variable overrides the default. Revenue cannot fall below zero (`monthly_revenue >= 0`); Volume loss cannot be negative (`volume_loss >= 0`); Cannot lose more than 100% of volume (`volume_loss <= 1`); Variable cost ratio cannot be negative (`variable_cost_ratio >= 0`).
- Time horizon: no horizon was supplied, so the model used the default for “raise prices, or hold” decisions, 24 months. The horizon chosen can materially affect the result, and for some decisions it decides the answer outright. Set it explicitly to the period you actually care about.

01 THE SHORT VERSION

What the result means

Why

Raise prices 8% delivers \$231,759 more than Hold prices on average. Its mean cumulative net cash flow over 24 months is \$1,101,972. Both options face the same simulated conditions, so this is a like-for-like comparison rather than two separate forecasts.

Range of gains

In 80% of simulated scenarios Raise prices 8% comes out ahead by between \$31,901 and \$402,496. It is behind in 7% of scenarios, and the largest simulated disadvantage is \$180,103.

Each option's own outcome spans a wide range (P10 to P90, worked out before rounding) — \$1,054,531 for Raise prices 8% and \$974,468 for Hold prices — but the gap between them moves far less: \$370,594. Most of the uncertainty is about the business, not about which option is better, and it affects both options together.

What rests on an estimate

This result depends partly on an estimated assumption: **Volume loss**. Use the flip-point analysis below to see how much that assumption would have to change before the comparison reverses.

A statement about this model, not advice. What to do about it is a judgement that has to account for things the model does not contain.

Before comparing this with anything: No one-time costs of making the change: telling customers, amending contracts and reworking billing are not included, only the ongoing revenue effect. A one-time cost of that kind can be the size of the difference reported above, so weigh it alongside this result rather than after it.

The assumption this result rests on

Volume loss estimated

The assumption closest to changing this result.

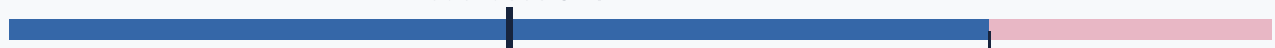
Model uses

5.4%

Result changes above

9.5%

Model uses 5.4%



Range entered: 1.0% to 12.0%

■ Raise prices 8% ahead ■ Hold prices ahead

Next step

Measure volume loss: both its typical level and how widely it could vary.

If its typical level could be above 9.5%, the comparison changes. If not, the result does not rest on it as heavily as the ranking suggests.

The assumption closest to flipping the comparison is **Volume loss**. The model currently uses **5.4%** — the value half of the simulated draws fall below (the median), not the most-likely figure of 4.0%, because the range runs from 1.0% to 12.0%, which is lopsided. The comparison flips once that median moves above **9.5%**.

Key metrics

Raise prices 8%

93%

of simulated scenarios

58% chance of cumulative net cash flow over
24 months reaching \$1,000,000

Hold prices

7%

of simulated scenarios

35% chance of cumulative net cash flow over
24 months reaching \$1,000,000

Compared over 100,000 simulated scenarios in which both options produced a usable result, across a 24-month horizon.

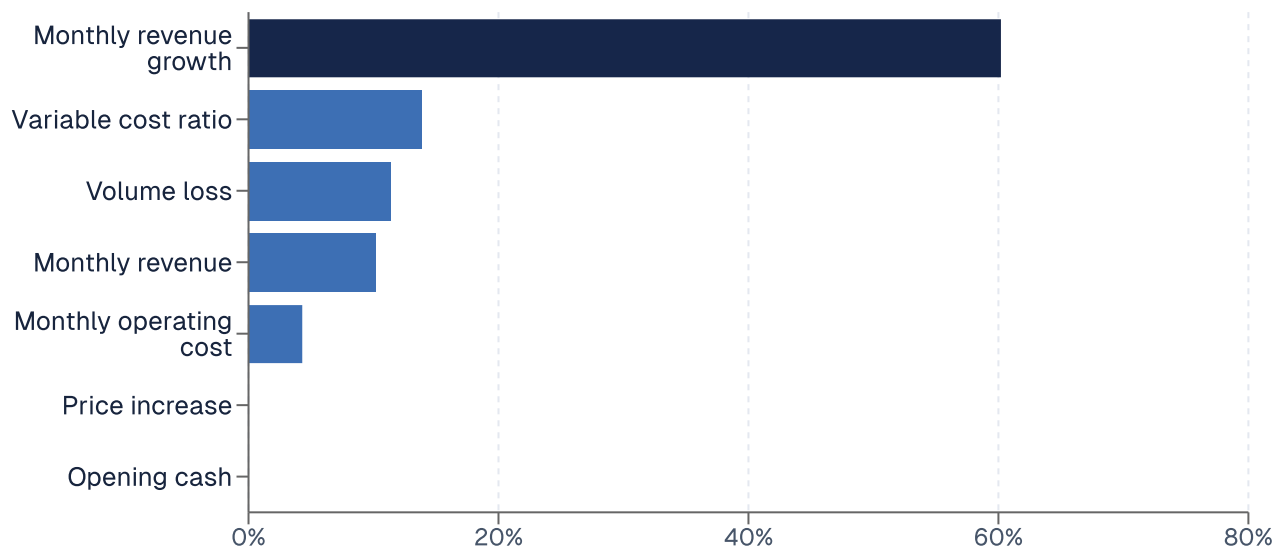
The large figure compares the two options with each other in the same scenario; the chance of reaching the target compares each option with the target on its own. They can differ: winning the comparison often does not mean clearing the target often, when both options tend to land on the same side of it.

Outcomes in detail

All figures below are cumulative net cash flow over 24 months.		
Metric	Raise prices 8%	Hold prices
Mean average across all scenarios	\$1,101,972	\$870,213
Median (P50) half land above, half below	\$1,085,840	\$853,353
P5	\$460,263	\$279,950
P10 downside	\$586,973	\$395,745
P90 upside	\$1,641,504	\$1,370,214
P95	\$1,799,443	\$1,514,524
Differences and ranges in this report are calculated before rounding, so each can sit \$1 from the difference of the figures shown.		
P10 to P90 contains the central 80% of simulated outcomes; P5 to P95 contains the central 90%.		
How precisely was the average measured?		
The percentile ranges above describe how uncertain the business outcome is. They do not narrow if you run more iterations — the world is not less uncertain for having been simulated more.		
Monte Carlo estimation precision. The estimated average has a 95% interval of \$2,517 for Raise prices 8% and \$2,321 for Hold prices. This measures simulation precision, not the uncertainty in the underlying business assumptions. It does shrink with more iterations, and if it is small beside the gap between the options, running more would not change the answer.		

03 INFLUENCE

What most influences the result



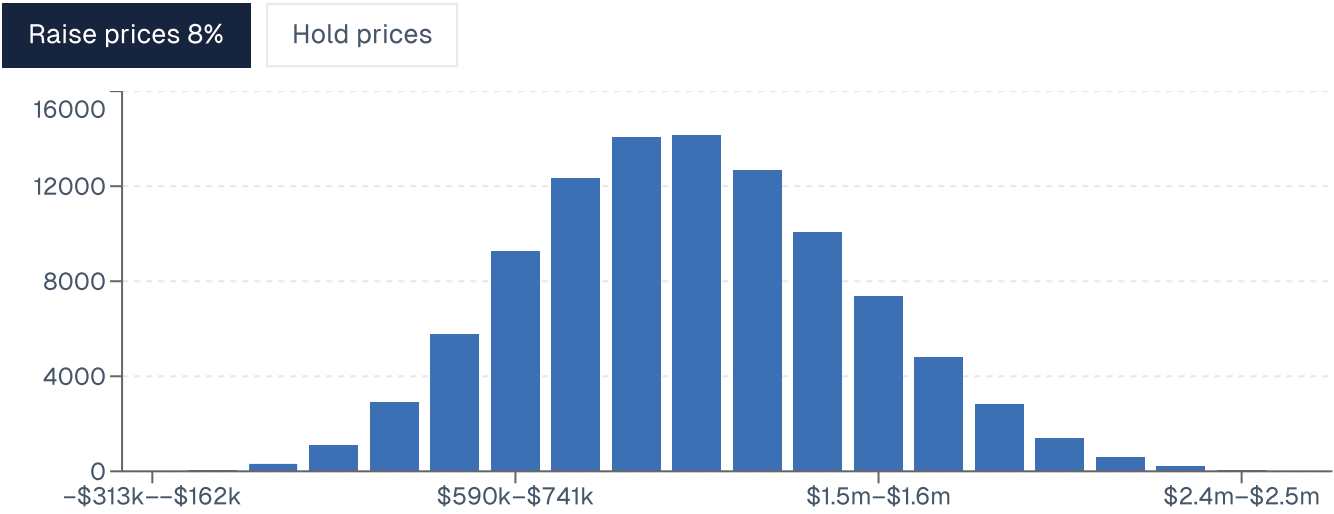
Price increase, Opening cash show 0% because they were entered as a single fixed value. A figure with no range cannot explain any of the variation in the outcome, however much it affects the decision — give it a low and a high if you want its influence measured.

Influence and the flip point answer different questions. This section shows what drives the variation in one option's own outcome. The flip-point analysis asks which assumption can change the comparison between the two options — so they can name different variables without disagreeing.

How influence is measured

Share of explained variance attributable to each assumption, by LMG (Shapley) decomposition. The shares are non-negative and sum exactly to 100% of what the assumptions explain between them, which is 99.8% of this option's outcome variance (R^2). Where assumptions are correlated the shared variance is split between them rather than counted twice. Association within the model, not causation.

Outcome distribution



Distribution of 100,000 valid simulated outcomes for **Raise prices 8%**.

Cash runway

Neither option runs out of cash in any simulated scenario. Cash is not the deciding factor here.

Raise prices 8%		Hold prices	
0%		0%	
chance of running out of cash		chance of running out of cash	
Lowest balance (P10)	\$523,742	Lowest balance (P10)	\$489,639
Extra opening cash for a 5% breach risk	none needed	Extra opening cash for a 5% breach risk	none needed

Opening balance \$520,000, floor \$0. Raise prices 8%'s lowest balance is above the opening, which means it was cash-positive from the first month.

Lowest balance (P10): in nine scenarios out of ten the lowest month-end balance stayed above this figure. Opening cash is counted in these balances but not in the cumulative net cash flow totals elsewhere in this report. Extra opening cash for a 5% breach risk: the amount to add to the opening balance so that no more than 5% of scenarios breach the floor; “none needed” means 5% or fewer already do.

No minimum cash level was entered, so this tests only whether cash reaches zero. Most businesses need a buffer above that for payroll, tax and lender requirements; enter the minimum you need to hold to test against it.

No month-by-month chart: the balance stays above the floor in every simulated scenario under both options, so there is no curve to draw. Plotting a flat zero would give it an axis and imply a risk the model did not find.

06 FLIP-POINT ANALYSIS

What would change the decision

The comparison (Raise prices 8%) rests on Volume loss. Its median (the value half of the simulated draws fall below) is 5.4%; the result changes to Hold prices once that median goes above 9.5%. That variable is flagged as an estimate with no supporting data, so both its current value and the range around it are guesses. The distance to the flip point therefore cannot be read as safety margin, and this result should be treated as unsettled until the figure is measured. Doing so is the single highest-value action before committing.

Assumption	Median now	Flips at	How close
Volume loss estimated	5.4%	above 9.5%	0.67 spreads
Price increase [†]	8.0%	below 4.3%	2.30 spreads
Monthly revenue [†]	\$248,252	nothing in the range searched	—
Monthly revenue growth	1.5%	nothing in the range searched	—
Variable cost ratio	21.3%	nothing in the range searched	—
Monthly operating cost	\$196,254	nothing in the range searched	—
Opening cash [†]	\$520,000	nothing in the range searched	—

[†] **Price increase.** This variable has no uncertainty of its own (it is specified as a single value), so its fragility is measured against a nominal 20% of its magnitude rather than a modelled spread.

[†] **Monthly revenue.** Moved far enough down, this assumption makes the two options identical. That is not a reversal: there is no longer a difference between them for either to win.

[†] **Opening cash.** This variable has no uncertainty of its own (it is specified as a single value), so its fragility is measured against a nominal 20% of its magnitude rather than a modelled spread.

“Median now” is the value half of each assumption’s simulated draws fall below. Where the range is lopsided it differs from the most-likely value.

What does “How close” mean?

For **Volume loss**: $(9.536\% - 5.367\%) \div 6.217\% = 0.67$ spreads. The spread is that assumption’s own 10th-to-90th-percentile width. These figures carry more decimal places than the table above, which rounds for reading; the ratio is computed before any rounding, so recomputing it from the figures in the table lands close but not exact.

It shows how far the current assumption is from the point where the comparison flips, measured relative to the uncertainty assigned to that assumption — so dollars, percentages and counts are

comparable. Smaller numbers mean the result is more sensitive to that assumption; below about one spread it is precarious.

The flip point is the median of the shifted distribution, not a single value to compare one measurement against: the variable stays uncertain on both sides of the line. Searched at 20,000 iterations per run, fewer than the headline simulation — enough to resolve which side of one half the scenario count falls on, which is all a flip point needs.

07 BOUNDARIES

Outside this model

These may affect the decision but are not quantified here. If any is material, add it as an assumption or weigh it alongside this result.

- Competitor response is not modelled. If a price rise triggers a competitor move, the volume loss is larger than whatever you entered.
 - One price change, applied once. Staged increases are not represented, and neither is a ladder of alternatives — this compares one increase against holding.
 - No effect on new-customer acquisition — only on the existing base.
 - No segment differences: the increase and the loss apply uniformly across all revenue.
 - No one-time costs of making the change: telling customers, amending contracts and reworking billing are not included, only the ongoing revenue effect.
 - The price rise and the volume loss are entered separately. The model does not make a larger increase cause a larger loss, so test a different increase by changing both.
 - Cash flows are not discounted, and there is no financing: if you would draw on a facility when cash got tight, the chance of breaching the floor is pessimistic.
 - No correlations between assumptions were declared: each was sampled independently of the others.
-

08 THE OTHER SIDE

Typical profile of scenarios where Hold prices wins

No assumption stands out in the scenarios where Hold prices wins: in those scenarios each one sits close to its usual value, so they are won by several small differences together rather than by one.

This describes the scenarios where Hold prices wins, taken together. The flip-point analysis above is a different test: it moves one assumption at a time until the result reverses. The two can look different without contradicting each other.

Where these numbers come from

1 assumption was estimated, not supplied

Volume loss — low confidence

Check these first. Every other figure traces back to a value you supplied.

Simulation run

e4652a2d-c82f-46a0-b158-fb2c8f2be3d3 · 100,000 simulations, P5–P95 outcome band, engine v1.39.0, seed 20260907

Model version

example-price-change-001 v1 · Model v1

Variable

v1 · Monthly revenue — Triangular(235000, 248000, 262000)

v2 · Price increase — Fixed(0.08)

v3 · Volume loss — Triangular(0.01, 0.04, 0.12)

v4 · Monthly revenue growth — Triangular(0.004, 0.014, 0.026)

v5 · Variable cost ratio — Triangular(0.16, 0.21, 0.27)

v6 · Monthly operating cost — Triangular(188000, 196000, 205000)

v7 · Opening cash — Fixed(520000)

Source

v1.source · Fitted from historical data

v2.source · Supplied by the user

v3.source · ESTIMATED — no supporting evidence

v4.source · Fitted from historical data

v5.source · Fitted from historical data

v6.source · Fitted from historical data

v7.source · Supplied by the user

Technical appendix

Everything needed to check or reproduce this result. Raise prices 8% vs hold — Kestrel Software · model example-price-change-001 v1 · run e4652a2d-c82f-46a0-b158-fb2c8f2be3d3 · 10/5/2026, 9:57:08 AM

1. The decision

Question	Raise prices 8% vs hold — Kestrel Software
Option A	Raise prices 8%
Option B	Hold prices
Objective	Mean cumulative net cash flow over 24 months, maximize
Target	Probability of cumulative net cash flow over 24 months reaching \$1,000,000
Horizon	24 months
Currency	USD

2. Assumptions

Variable	Distribution	Applies to	Source
Monthly revenue	Triangular, \$235,000 to \$262,000, most likely \$248,000	Both	Fitted from historical data (high confidence) — Billed MRR, trailing 12 months.
Price increase	Fixed at 8.0%	Option A	Supplied by the user (high confidence)
Volume loss	Triangular, 1.0% to 12.0%, most likely 4.0%	Option A	ESTIMATED — no supporting evidence (low confidence) — No historical basis. Kestrel has never raised prices. FLAGGED.
Monthly revenue growth	Triangular, 0.4% to 2.6%, most likely 1.4%	Both	Fitted from historical data (medium confidence)
Variable cost ratio	Triangular, 16.0% to 27.0%, most likely 21.0%	Both	Fitted from historical data (medium confidence) — Hosting, support, payment processing.
Monthly operating cost	Triangular, \$188,000 to \$205,000, most likely \$196,000	Both	Fitted from historical data (high confidence)
Opening cash	Fixed at \$520,000	Both	Supplied by the user (high confidence)

3. Constraints and guardrails

Monthly revenue ≥ 0 — Revenue cannot fall below zero (applied automatically by this template)

Volume loss ≥ 0 — Volume loss cannot be negative (applied automatically by this template)

Volume loss ≤ 1 — Cannot lose more than 100% of volume (applied automatically by this template)

Variable cost ratio ≥ 0 — Variable cost ratio cannot be negative (applied automatically by this template)

Bound constraints clip the variable; relational constraints exclude the whole iteration. A constraint you declare on a variable replaces the template's own for that variable.

4. How the numbers were produced

4.1 Cash flow for this decision

```
realised[i,m] = clip( (m + 1) / churn_response_months[i] , 0 , 1 )

Raise:  price_factor = 1 + price_increase[i]
        volume_factor[i,m] = 1 - volume_loss[i] × realised[i,m]
Hold:   price_factor = 1 , volume_factor = 1

growth[i,m] = (1 + monthly_revenue_growth[i])^m
revenue[i,m] = monthly_revenue[i] × growth[i,m] × price_factor ×
volume_factor[i,m]
var_cost[i,m] = monthly_revenue[i] × variable_cost_ratio[i]
                × growth[i,m] × volume_factor[i,m]

cash_flow[i,m] = revenue[i,m] - var_cost[i,m] - monthly_operating_cost[i]
```

The asymmetry is the whole economic case for raising prices: price_factor multiplies revenue but NOT variable cost, because the cost of serving a customer does not rise because you charged them more. Volume loss reduces both – you lose the revenue and stop incurring the cost of serving those customers.

4.2 Aggregation

```
value[i] = Σ(m=0..M-1) cash_flow[i,m]
```

Undiscounted. No time-value adjustment is applied.

4.3 Sampling and correlation

C	correlation matrix over the k variables, identity off the stated dependencies; nudged to the nearest positive semi-definite matrix by eigenvalue clipping if the stated pairs are jointly inconsistent
L	lower Cholesky factor of C, so $L L^T = C$
$z[i]$	$\sim N(0, I_k)$ independent standard normals
$y[i]$	$= L z[i]$ correlated standard normals
$u_j[i]$	$= \Phi(y_j[i])$ to uniforms, Φ the standard normal CDF
$x_j[i]$	$= F_j^{-1}(u_j[i])$ to the variable's own distribution

F_j^{-1} is the inverse CDF (quantile function) of variable j's stated distribution. Because each variable is transformed by its own F_j^{-1} , every marginal distribution is exact regardless of the correlation applied – only the joint dependence comes from the copula. This is a Gaussian copula, and it attenuates the stated coefficient slightly: the correlation of the delivered samples is a little below the correlation of the normals, and more so for strongly skewed marginals.

4.4 Constraints

min_bound b on variable j:	$x_j[i] \leftarrow \max(x_j[i], b)$	clip, in place
max_bound b on variable j:	$x_j[i] \leftarrow \min(x_j[i], b)$	clip, in place
relational expression E:	$\text{valid}[i] = \text{valid}[i] \wedge E(x[i])$	exclude the row

Bound constraints censor: mass that would fall outside the bound piles up at it, which changes the variable's distribution. Relational constraints reject: the whole iteration is dropped from every statistic, which reduces the effective sample size rather than reshaping any variable.

4.5 Statistics

valid[i]	iteration i satisfied every constraint	
comparable[i]	valid[i] $\wedge$ value_a[i] and value_b[i] are both finite	
N_v, N_c	counts of the above	
expected value	$E[\text{value}] = (1/N_v) \sum(\text{valid}) \text{value}[i]$	
median	P_{50} , by linear interpolation between order statistics	
P_q	the q-th percentile of value over valid iterations	
P(target)	$(1/N_v) \sum(\text{valid}) 1[\text{value}[i] \geq \text{target}]$	(maximise)
	$(1/N_v) \sum(\text{valid}) 1[\text{value}[i] \leq \text{target}]$	(minimise)
win fraction	$w_a = (1/N_c) [\sum(\text{comparable}) 1[\text{value}_a[i] > \text{value}_b[i]] + \frac{1}{2} \sum(\text{comparable}) 1[\text{value}_a[i] = \text{value}_b[i]]]$	
standard error	$SE = s / \sqrt{N_v}$, s the sample standard deviation of value	
95% estimation		
interval	$\pm 1.95996 \times SE$	– on the ESTIMATE of the mean, not on the outcome. It shrinks as $1/\sqrt{N}$; the percentiles above do not shrink at all.

The win fraction is a PAIRED comparison: options are compared within the same simulated future, on the same draws. Uncertainty common to both options therefore cancels out of it, which is why a win probability can be extreme while the outcome ranges overlap almost entirely.

4.6 Sensitivity

LMG (Lindeman, Merenda and Gold), the Shapley decomposition of R^2 . For each variable, the average increase in R^2 it produces when added to the model, averaged over every ordering of the variables:

$$\text{share}_j = \frac{\sum_{S \subseteq P \setminus \{j\}} |S|! (m - |S| - 1)!}{m!} \times [R^2(S \cup \{j\}) - R^2(S)]$$

where P is the set of variables applying to the winning option, $m = |P|$, and

$$R^2(S) = \mathbf{r}_S^T \mathbf{R}_{SS}^{-1} \mathbf{r}_S$$

computed from the correlation matrix rather than by refitting, so the cost depends on the number of VARIABLES, not the number of iterations. Reported as a percentage of the total explained variance.

This is the unique attribution with the properties a decomposition needs: the shares are non-negative, they sum exactly to the model's R^2 , and a variable's share does not depend on the order the analyst happened to list them in. Where two variables are correlated the variance they share is SPLIT between them rather than counted against both.

Above twelve variables the exact computation (2^m subsets) becomes unreasonable and the engine falls back to normalised marginal R^2 :

$$\text{contribution}_j = 100 \times R^2_j / \sum_k R^2_k, \quad R^2_j = \text{corr}(x_j, \text{value_winner})^2$$

which double-counts shared variance – the raw sum exceeded 1 on a reference model, and dividing by it forces the printed figures to total 100% without removing the overlap. When the fallback is used the Result says so, in `sensitivity_method` and in a warning; read the ranking rather than the percentages in that case.

Association within the simulations, not causation.

4.7 Reverse sensitivity

$$L = \{ i : \text{comparable}[i] \wedge \text{the losing option won in future } i \}$$

For each variable j applying to the losing option:

$$\begin{aligned} \text{pop}_j &= \text{median}(x_j[i] : \text{comparable}[i]) \\ \text{cond}_j &= \text{median}(x_j[i] : i \in L) \\ \text{spread}_j &= P_{90}(x_j) - P_{10}(x_j) \quad \text{over comparable iterations} \\ \text{strength} &= |\text{cond}_j - \text{pop}_j| / \text{spread}_j \end{aligned}$$

Reported only if $\text{strength} \geq 0.10$.

These are conditional medians, not thresholds. The filter exists because a variable applying to BOTH options cancels from the comparison and cannot be a condition for either winning, yet its conditional median still differs from the population median by sampling noise – which was published as a finding until the filter was added.

4.8 Threshold search

For each variable j , shift its whole distribution by δ , preserving shape and spread, and re-run:

$$F_{j,\delta^{-1}}(u) = F_j^{-1}(u) + \delta \quad (\text{additively, for location-family variables})$$

The shifted median is $\text{median}_j + \delta$. Define

$$g(\delta) = w_a(\text{model with variable } j \text{ shifted by } \delta) - 0.5$$

and find δ^* with $g(\delta^*) = 0$ by expanding a bracket geometrically, then bisectioning. The reported flip point is $\text{median}_j + \delta^*$, and

$$\text{fragility}_j = |\delta^*| / \text{spread}_j$$

so distances are comparable across variables measured in different units.

Shifting the whole distribution, rather than only its central value, answers "what if this quantity were higher, with the same uncertainty around it" rather than confounding a change in location with a change in shape. The flip point is therefore the MEDIAN of a shifted distribution – not a single value to compare one measurement against, since the variable remains uncertain on both sides of it.

The search assumes w_a moves monotonically in δ . Every shipped template is monotone in every variable, verified by direct sweep; a future template that is not would return one crossing of several.

4.9 Cash balance and probability of ruin

```
balance[i,m] = opening_cash[i] +  $\Sigma(t=0..m)$  cash_flow[i,t]
floor[i]      = minimum_cash[i], or 0 if not supplied
ruined[i]     = 1 if  $\min(m)$  balance[i,m] < floor[i]
```

```
P(ruin)                = (1/N_v)  $\Sigma(\text{valid})$  ruined[i]
P(ruin by month m)     = (1/N_v)  $\Sigma(\text{valid})$  1[  $\exists t \leq m : \text{balance}[i,t] < \text{floor}[i]$  ]
```

Ruin is breaching the floor at ANY period, not ending below it: a business that runs out in month 9 has run out, whatever its closing balance says. A cumulative total cannot show this, which is the reason the path is modelled.

Computed only where the template produces a complete net cash flow AND every component is supplied. Otherwise the engine declines, because a ruin probability built from a partial cash flow is reassuring and wrong.

5. Results

Statistic	Raise prices 8%	Hold prices
Mean cumulative net cash flow over 24 months	\$1,101,972	\$870,213
Median (P50) cumulative net cash flow over 24 months	\$1,085,840	\$853,353
P10 cumulative net cash flow over 24 months	\$586,973	\$395,745
P90 cumulative net cash flow over 24 months	\$1,641,504	\$1,370,214
95% Monte Carlo estimation interval on the mean	±\$2,517	±\$2,321
95% Monte Carlo interval on the scenarios-favouring figure	±0.15 percentage points, over 100,000 compared scenarios	
Probability of cumulative net cash flow over 24 months reaching \$1,000,000	58%	35%

The percentiles describe how uncertain the outcome is and do not narrow with more iterations. The estimation interval describes how precisely this run measured the average and does. They are different quantities.

Run diagnostics

Scenarios simulated	100,000
Valid after constraints	100,000
Comparable for both options	100,000 (the denominator of every comparison)
Tied scenarios	none
Influence method	LMG (Shapley) decomposition; the assumptions explain $R^2 = 99.8\%$ of the favoured option's variance
Influence shares	Monthly revenue growth 60.2%; Variable cost ratio 13.9%; Volume loss 11.4%; Monthly revenue 10.2%; Monthly operating cost 4.3% (shares of the 99.8% of outcome variance the assumptions explain, not of the total)
Correlations	none declared: every assumption was sampled independently
Extra opening cash for a 5% breach risk	Raise prices 8%: none needed; Hold prices: none needed

6. Defaults and adjustments applied by the engine

Template defaults

Time for customers to react: not supplied. Defaults to 3 months.

Minimum acceptable cash: not supplied. Omitted: the floor is treated as zero, so a breach is only reported once cash has actually run out — later than most businesses can afford to find out.

Each default is a single fixed value used in every scenario; it is not sampled, so it has no range, no influence share and no flip point.

Warnings

Model constraints: 4 constraint(s) required for “raise prices, or hold” decisions were applied automatically. A constraint you declare on the same variable overrides the default. Revenue cannot fall below zero ($\text{monthly_revenue} \geq 0$); Volume loss cannot be negative ($\text{volume_loss} \geq 0$); Cannot lose more than 100% of volume ($\text{volume_loss} \leq 1$); Variable cost ratio cannot be negative ($\text{variable_cost_ratio} \geq 0$).

Time horizon: no horizon was supplied, so the model used the default for “raise prices, or hold” decisions, 24 months. The horizon chosen can materially affect the result, and for some decisions it decides the answer outright. Set it explicitly to the period you actually care about.

7. Provenance chain

`simulation e4652a2d-c82f-46a0-b158-fb2c8f2be3d3` — 100,000 simulations, P5–P95 outcome band, engine v1.39.0, seed 20260907

`model_version example-price-change-001 v1` — Model v1

`horizon objective.horizon_months` — 24 months (template default)

`variable v1` — Monthly revenue — Triangular(235000, 248000, 262000)

`variable v2` — Price increase — Fixed(0.08)

`variable v3` — Volume loss — Triangular(0.01, 0.04, 0.12)

`variable v4` — Monthly revenue growth — Triangular(0.004, 0.014, 0.026)

`variable v5` — Variable cost ratio — Triangular(0.16, 0.21, 0.27)

`variable v6` — Monthly operating cost — Triangular(188000, 196000, 205000)

`variable v7` — Opening cash — Fixed(520000)

`source v1.source` — Fitted from historical data

`source v2.source` — Supplied by the user

`source v3.source` — ESTIMATED — no supporting evidence

`source v4.source` — Fitted from historical data

`source v5.source` — Fitted from historical data

`source v6.source` — Fitted from historical data

`source v7.source` — Supplied by the user

`default churn_response_months` — Time for customers to react — not supplied, template default applied. Defaults to 3 months.

`default minimum_cash` — Minimum acceptable cash — not supplied, template default applied. Omitted: the floor is treated as zero, so a breach is only reported once cash has actually run out — later than most businesses can afford to find out.

8. Reproducibility

Simulation run ID	e4652a2d-c82f-46a0-b158-fb2c8f2be3d3
Model	example-price-change-001 version 1
Engine version	1.39.0
Random seed	20260907
Iterations	100,000
Runtime	0.33s

Re-running this model at seed 20260907 on engine v1.39.0 reproduces every figure in this document exactly.